Mobile Architectures

Practical Work – *Android* - *Kotlin* – 2024/2025

In the context of this work, the intention is to develop an *Android application*, called *Quizec*, which allows questionnaires to be used in the classroom. The teacher will be able to create questionnaires that include various types of questions, for the students to answer.

All users of the application will be able to create questionnaires. The user who will generate and share a questionnaire with other users is designated as the "*creator*". Once created, the questionnaire will receive a unique identifier (alphanumeric with 6 characters), and a descriptive text and, optionally, an image (optional from the user's point of view, but the application must provide this option). A questionnaire consists of a sequence of questions, which can be reused in different questionnaires from the same creator.

The questions to be created must also have a unique identifier, a title, an image (optional from the user's point of view, but the application must provide this option), and can be of the following types (the following list also provides information with *suggestions* for ways to present the results):

- *P01 – Yes/No, True/False questions.*
  - Results: number of responses; bar graph, pie chart.
- *P02 – Multiple choice questions with only one correct answer (2-6 options)*
  - Results: number of responses; bar graph, pie chart.
- *P03 – Multiple choice questions with more than one correct answer (2-6 options)*
  - Results: number of responses; bar graph, pie chart.
- *P04 – Matching questions, where students must match items from one column with corresponding items from another column (2-6 options).*
  - Results: list sorted in descending order of most chosen options
- *P05 – Ordering questions, where students must place a series of items in: chronological order, hierarchical order, etc. (2-6 options)*
  - Results: list sorted in descending order of most chosen options
- *P06 – Fill-in-the-blank questions, where students complete a sentence, equation, or text with the correct word or term (the list of possible options is given).*
  - Results: list sorted in descending order of most chosen options
- *P07 – Association questions, where students must associate an image or concept with a description or definition.*
  - Results: list sorted in descending order of most chosen options
- *P08 – Questions with answers based on filling-in missing words (individual boxes for free introduction of each word).*
  - Results: list sorted in descending order of most chosen options

A user can create as many questions as they want and save them for future use.

When the *creator* intends to use one of the quizzes in class, he/she provides the identifier to the students and can set a maximum time for the answer. The *creator* also defines:

- whether students have immediate access to the quiz (after providing the quiz ID) or only when the *creator* authorizes it (to allow all users to connect before viewing). In this second situation, there should be a waiting screen that properly informs users already connected;
- whether an operating mode is activated which restricts responses only to users who are in a geographic location close to where the questionnaire was made available (location of the *creator*);
- whether students have immediate access to the results right after their answer or only at the end of the defined time. The *creator* will have immediate access to the results that are being entered right after opening the questionnaire.

The *creator* can end the time whenever he wants. For example, he/she can check that all connected users have already responded, without having to extend the questionnaire for the entire time initially set.

All users must be authenticated to ensure that each user only answers a survey once. However, it must be ensured that users (including the *creator*) do not have access to personal information on other users' responses – they only have access to the anonymous aggregated results.

The *creator* can also:

- Manage question database: duplicate, edit and delete questions (it must be ensured that editing and deletion do not affect already completed questionnaires);
- Edit or delete a questionnaire that has not yet been made available to other users;
- Duplicate a questionnaire. When duplicating a questionnaire that is already available and has answers, the status information and answers will not be copied and can be edited as indicated above.

Any user must have access to their history of questionnaires in which they have participated, and can duplicate questions from any of them, acting as the *creator* for that copy.

Please note that duplicate questions and questionnaires should have a unique new identifier.

Notes on development

Application development must be carried out by using *Jetpack Compose*.

All information should be stored using services that facilitate sharing between users. It is recommended to use *Firebase* services, but another similar service (e.g. *Appwrite*, *Supabase* or *Parser*) can also be chosen. Considering the implementation using *Firebase*, storage can be done using the following services: *Firestore* or *Storage* (e.g. to save images). The application should require user registration and authentication to ensure proper functioning of the application, for example, to associate the user with a created questionnaire or to ensure that a user only responds once.

Define a user interface that is appropriate for the program's objectives, and that is intuitive and pleasant. The application should have English as the default language, and should support additional languages, at least Portuguese. The language to be displayed is the one configured on the device. The text inside the questionnaire and individual questions are not subject to automatic translation. The application should be able to run in *Portrait* or *Landscape* orientation without losing functionality.

The internal organization of the application, code quality and robustness will be important elements in the evaluation.

Groups: 3 elements (max)

Delivery date: 23:59 on 2025.01.05
Warning: 5% will be deducted for each hour of delay

Defense: The defense is mandatory and will take place on the 6th, 7th and 8th of January (in *a slot* to be scheduled through *Nónio*)

Delivery method:
Delivery of a single file in **ZIP** format via *Nónio* with the following name:

`AMOV2425.<nr_student1>.<nr_student2>.<nr_student3>.zip`

- **Files in other formats will be subject to a penalty of up to 5% in the final grade**

This file should include:

- all the code (folders with the projects) with all the essential resources for compilation and execution. In *Android Studio* projects, the following directories must be removed beforehand: `<proj>/.gradle` and `<proj>/app/build`
  - **If the files mentioned are not removed, a penalty of up to 5% will be applied to the final grade**
- technical report (PDF)
- Poster about the application (PDF): minimum format A2 and, in addition to images and short sentences that highlight the application's features, it must include information about ISEC-DEIS, the curricular unit, students in the group and the date
  - The poster will be evaluated as a bonus in the work, with a maximum grade of 5%, but only if the work is minimally functional.